SON N.T. TU

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Research Interests: Hamilton–Jacobi equations, free boundary problems, integro-differential equations, optimal control, homogenization, dynamical systems

Academic Appointments

- *Assistant Professor* — Aug 2025 – Present
  Department of Mathematics, Baylor University, Waco, TX
- *Visiting Assistant Professor* — Aug 2022 – Jul 2025
  Department of Mathematics, Michigan State University (MSU), East Lansing, MI
- *Teaching Assistant* — 2015 – 2016
  Vietnam National University, Ho Chi Minh City (VNU-HCMC), Vietnam

Education

- Mathematics Ph.D., University of Wisconsin-Madison (UW-Madison). Advisor: Hung Tran — Aug 2022
- B.S. Honors program in Mathematics, University of Science, VNU-HCMC — Aug 2015

Publications and Preprints

Preprints and Submitted Manuscripts

17. **Son N. T. Tu**, Jianlu Zhang, Siyao Zhu. "Convergence Rate of Birkhoff Average for Toral Quasi-Periodic Rotations and Applications." *Submitted (2026)*
16. Tan-Phuc Nguyen, Minh-Binh Tran, **Son N. T. Tu**. "A Structure-Preserving Neural-Spectral Method for Reconstructing Controls of Wave Equations." *Submitted (2026)*
15. Gengyu Liu, **Son N. T. Tu**, Jianlu Zhang. "Quantitative homogenization for static contact Hamilton-Jacobi equations". *Submitted (2026) – under minor revision at "Indiana University Mathematics Journal"*
14. Alain Bensoussan, Thien P.B. Nguyen, Minh-Binh Tran, **Son N. T. Tu**. "Operator Splitting, Policy Iteration, and Machine Learning for Stochastic Optimal Control". *Submitted (2026)*
13. Yuxi Han and **Son N. T. Tu**. "Quantitative homogenization of Hamilton–Jacobi equations on perforated domains with Dirichlet boundary conditions". *Submitted (2025)*

Published

12. Russell Schwab, **Son N. T. Tu**, and Olga Turanova. "Well-posedness for viscosity solutions of the one-phase Muskat problem in all dimensions". *Journal of Functional Analysis, Volume 292, Issue 2, (Jan 2027), 111667*
11. **Son N. T. Tu** and Jianlu Zhang. "Vanishing discount limits for first-order fully nonlinear Hamilton-Jacobi equations on noncompact domains". *Advances in Continuous and Discrete Models, Vol 2026, 58 (2026)*
10. Prerona Dutta, Khai T. Nguyen, and **Son N. T. Tu**. "On the rate of convergence in superquadratic Hamilton–Jacobi equations with state constraints". *Nonlinear Differential Equations and Applications NoDEA, Vol 33, 63, (2026)*
9. Bingyang Hu, **Son N. T. Tu**, and Jianlu Zhang. "Polynomial convergence rate for quasiperiodic homogenization of Hamilton–Jacobi equations". *Communications in Partial Differential Equations 50, no. 1–2 (February 1, 2025): 211–244.*

8. **Son N.T. Tu** and Jianlu Zhang. "On the regularity of stochastic effective Hamiltonian". *Proceedings of the American Mathematical Society 153 (2025), pp. 1191-1203.*

7. **Son N.T. Tu** and Jianlu Zhang. "Generalized convergence of solutions for nonlinear Hamilton–Jacobi equations with state-constraint". *Journal of Differential Equations 406 (Oct. 2024), 87-125.*

6. Farid Bozorgnia, Dohyun Kwon, and **Son N.T. Tu**. "The regularity with respect to domains of the additive eigenvalues of superquadratic Hamilton–Jacobi equation". *Journal of Differential Equations, 402, (Sep. 2024), 518-553.*

5. Yuxi Han and **Son N.T. Tu**. "Remarks on the vanishing viscosity process of state-constraint Hamilton–Jacobi equations". *Applied Mathematics & Optimization, 86(3) (Jun. 2022).*

4. **Son N.T. Tu**. "Vanishing discount for Hamilton–Jacobi equation in nested domains". *Journal of Differential Equations, 317, (Apr. 2022), 32-69.*

3. Yeon-Eung Kim, Hung Vinh Tran, and **Son N.T. Tu**. "State-constraint static Hamilton–Jacobi equations in nested domains". *SIAM Journal on Mathematical Analysis, 52(5) (Sep. 2020), 4161–4184.*

2. **Son N.T. Tu**. "Rate of Convergence for Periodic Homogenization of Convex Hamilton–Jacobi Equations in One Dimension". *Asymptotic Analysis, 121(2) (Jan. 2021), 171–194.*

Refereed conference proceedings & papers

1. Thu Nguyen, Quang M. Le, **Son N.T. Tu**, and Binh Nguyen. "Unequal Covariance Awareness for Fisher Discriminant Analysis and Its Variants in Classification". *2022 International Joint Conference on Neural Networks (IJCNN), (Jul. 2022).*

Grants, Honors, and Awards

- *AMS-Simons Travel Grant* award 2026 — Jul 2026 – Jul 2028
- *Research Travel Support from the Office of Postdoctoral Affairs*, MSU — 2024
- *2023–2024 Postdoctoral Prize for Excellence in Teaching*, Department of Mathematics, MSU — 2024
- *Teaching Assistant Superior Rating* — FA 2017, FA 2018, FA 2019, FA 2020
  Mathematics Department, UW-Madison
- *Graduate Research Travel Grant*, Graduate School, UW-Madison — 2021
- *GSSC Fellowship*, Graduate School, UW-Madison — 2021
- *Excellence in Research Award*, Mathematics Department, UW-Madison — 2020
- *Outstanding Teaching Assistant Award*, Mathematics Department, UW-Madison — 2020
- *Vietnam Education Foundation (VEF) Fellowship* (declined) — 2016
- *Valedictorian Award*, University of Sciences, VNU-HCMC, Vietnam — 2015
- *Third prize, Vietnam Mathematical Olympiad* (VMO) — 2011
- *World Finalist*, Shing-Tung Yau High School Mathematics Awards, Beijing, China — 2010

Services Contributions & Outreach

Organizing

- Co-organizer: *Summer Meeting 2026*. Saigon University (SGU), Vietnam — Jun 27, 2026
- Organizer: *Intensive Research Collaboration Program (IRCP 2026)* — Jun 01-09, 2026
  Organizer: *Workshop: Recent Developments in Hamilton–Jacobi Equations* — Jun 08-09, 2026
  Vietnam Institute for Advanced Study in Mathematics (VIASM), Hanoi, Vietnam
- Co-organizer: *Special Session on Advances and challenges in the study of nonlinear PDEs* — Oct 01-03, 2025
  AMS 2025 Fall Southeastern Sectional Meeting, Tulane University, New Orleans, LA, USA
- Co-organizer: — Mar 29-30, 2025
  Special Session on Frontiers in Nonlinear PDEs and Applied Mathematical Challenges
  AMS 2025 Spring Central Sectional Meeting, University of Kansas, Lawrence, KS, USA
- Co-organizer: *Madison PDEs Conference*, UW-Madison
  Originally scheduled for April 2020; canceled due to COVID-19
- Co-organizer: *AMS Student Chapter Seminar*, UW-Madison — 2018–2019

University Service – Baylor University

- Faculty Advisor, Möbius Mathematics Society — Sep 2025 – Present
  Advised student initiatives in learning and outreach, Baylor University.
- Committee Member, Baylor Math Circle — Sep 2025 – Present
  Support a middle school outreach program promoting interest and confidence in mathematics and STEM.
- Member, Ph.D. Dissertation Defense Committee (Mathematics) — May 25, 2026
- Session Chair, Faculty Contributed Talks, 105th Annual Meeting of the Texas Section Mathematical Association of America — Mar 27–28, 2026
- Invitation to Excellence – Friday Evening Reception, Baylor University — Feb 06-07, 2026

Service

- Guest Editor: Special Issue — 2026
  Topical Collection Optimal Control, Mean-Field Games, and PDEs in Decision Models
  Advances in Continuous and Discrete Models (ACDM)
- Referee for Mathematics Journals:
  - *Calculus of Variations and PDEs · Transactions of the American Mathematical Society* — 2026
    SIAM Journal on Mathematical Analysis (SIMA, 3 reviews)
    Advances in Continuous and Discrete Models (ACDM, 2 reviews)
  - *Mathematical Control and Related Fields · Advances in Continuous and Discrete Models (ACDM)* — 2025
    Journal of Differential Equations (JDE) · MathScinet
  - *Proceedings of the American Mathematical Society* — 2024
  - *Journal of Mathematical Physics · Journal of Geometric Analysis* — 2023
    Discrete and Continuous Dynamical Systems (DCDS)

Community Service

- Texas Section Project NExT Fellow, Mathematical Association of America (MAA) — 2025 – 2026
  Program supporting early-career faculty in undergraduate mathematics education.

- 20th Annual Texas Undergraduate Mathematics Conference (TUMC) Oct 18, 2025
  Co-led a team of seven Baylor undergraduates (Möbius Mathematics Society)
  to the conference at St. Mary's University, San Antonio.
- *Portal to the Public training* Mayborn Museum, Waco, Texas Sep 27, 2025
  Workshop on effective science communication and public engagement.
- Interactive STEM demonstration Mar 01, 2025
  for middle school students, as part of the *Girls Math and Science Day*, MSU
- Lead an interactive STEM demonstration table on "*Soap Bubbles and Minimal Surfaces*" Mar 09, 2024
  for middle school students, as part of the *Girls Math and Science Day*, MSU
- Judge for *University Undergraduate Research and Arts Forum 2023 (UURAF 2023)*, MSU Apr 14, 2023

Teaching

Ph.D. Student Supervision

1. Danie Paraiso Summer 2026 – Present

Undergraduate Research Mentoring

- *Undergraduate Research Mentor* for Minh Nguyen, MSU Summer 2024 – Summer 2025
  Uniqueness set for Hamilton-Jacobi equations with state-constraints
  Awarded *College of Natural Science Undergraduate Research Support Scholarship* for Summer 2024.
- *Directed Studies* (MTH490): Minh Nguyen, MSU Spring 2024
  Topic: *Optimal control theory and viscosity solutions to Hamilton–Jacobi equations*
  with *Best Presentation Award* at the *21st Math Student Conference*, MSU
- *Directed Reading Program:* William Robert Korbitz and Luanda Cai, UW-Madison Spring 2019
  Topic: *Optimal Control for Linear Systems*
- *Undergraduate PDEs Summer School:* Daotong Ge and Hangyu Pi, UW-Madison Summer 2017
  Co-mentored with Hung Tran

BAYLOR UNIVERSITY		Role	# St	Term
Calculus Of Variations	MTH 6V23	Instr. of Record	8	Fall 2026
Calculus II	MTH 1322	Instr. of Record	19	Fall 2026
Graduate Research	MTH 6V00 19	Instr. of Record	1	Fall 2025
Graduate Research	MTH 6V00 19	Instr. of Record	1	Summer 2025
Linear Algebra	MTH 2311	Instr. of Record	30	Spring 2026
Calculus I	MTH 1311	Instr. of Record	28	Fall 2025
Multivariable Calculus (Calculus III)	MTH 2321	Instr. of Record	18	Fall 2025
Graduate Research	MTH 6V00 19	Instr. of Record	1	Fall 2025

MICHIGAN STATE UNIVERSITY	Role	# St	Term
Topic in Optimal Control Theory, MTH 496-002 (Capstone course)	Instr. of Record	22	Spring 2025
Multivariable Calculus, MTH 234	Instr. of Record	191	Fall 2024
Matrix Algebra with Computational Applications, MTH/CMSE 314	Instr. of Record	30	Summer 2024
Directed Reading, MTH 490 (Introduction to Optimal Control Theory)	Instr. of Record	1	Spring 2024
Multivariable Calculus, MTH 234	Instr. of Record	170	Spring 2024
Multivariable Calculus, MTH 234	Instr. of Record	60	Fall 2023
Linear Algebra and Application to Data Science, MTH/CMSE 314	Instr. of Record	60	Spring 2023
Linear Algebra and Application to Data Science, MTH/CMSE 314	Instr. of Record	60	Fall 2022

UNIVERSITY OF WISCONSIN–MADISON	Role	# St	Term
College Algebra, Math 112	Instr. of Record	60	Fall 2021
College Algebra, Math 112	Instr. of Record	30	Spring 2021
Undergraduate PDE, Math 619	Teach. Asst.	$\sim$ 30	Spring 2021
Business Calculus, Math 211	Recitation Instr.	$\sim$ 30	Fall 2020
Mathematical Analysis I, Math 521	Teach. Asst.	$\sim$ 30	Summer 2020
College Algebra, Math 112	Recitation Instr.	$\sim$ 60	Fall 2019
Multi-variable Calculus, Algebra & Differential Equations, Math 375	Recitation Instr.	$\sim$ 30	Spring 2019
Multi-variable Calculus, Algebra & Differential Equations, Math 376	Recitation Instr.	$\sim$ 30	Fall 2018
Business Calculus, Math 211	Recitation Instr.	$\sim$ 60	Fall 2017
Linear Algebra & Differential Equations, Math 319	Recitation Instr.	$\sim$ 60	Spring 17
Multi-variable Calculus 2, Math 222	Recitation Instr.	$\sim$ 60	Fall 2016

VIETNAM NATIONAL UNIVERSITY, HCMC	Role	# St	Term
Calculus II, MATH2153 (Excellent Program - Univ. of Informatics)	Recitation Instr.	$\sim$ 60	Spring 2016
Calculus III, MATH253 (Adv. Comp. Sci. - Univ. of Science)	Recitation Instr.	$\sim$ 60	Fall 2015

Selected Presentations

Selected Invited talks

38. *PDE/Applied Math Seminar* — Nov 16, 2026
Indiana University - Bloomington, Department of Mathematics

37. *Advances in Theory, Applications and Numerics for PDE and Optimal Control* — Nov 7-8, 2026
2026 Fall Western Sectional Meeting, Arizona State University, Tempe, Tempe, AZ
Operator Splitting and Policy Iteration for Second-Order Hamilton–Jacobi Equations

36. *Special session on Free Boundary Problems and Regularity for PDEs* — Nov 7-8, 2026
2026 Fall Western Sectional Meeting, Arizona State University, Tempe, Tempe, AZ

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35. (Virtual) *ISQGD–SS16* — October 11, 2026
Special Session on Recreational Mathematics, Mathematical Games, Puzzles & Patterns
International Society for Quality Growth and Development (ISQGD).
Irrational Motion, Uniform Distribution, and Homogenization

34. *Homogenization in PDE and Stochastic Processes 2026* — Aug 24-26, 2026
Research Institute for Mathematical Sciences, Kyoto University (RIMS)
Quantitative Quasi-Periodic Homogenization through Birkhoff Averages

33. *Workshop on Pure and Applied Analysis 2026* — Aug 15-16, 2026
Texas A&M University, Department of Mathematics
Convergence Rate of Birkhoff Average for Toral Quasi-Periodic Rotations and Applications

32. *Workshop: Summer School and Workshop on Optimization and Decision Theory* July 16, 2026
International Centre for Interdisciplinary Science and Education (ICISE)
07 Science Avenue, Quy Nhon Nam Ward, Gia Lai, Vietnam.
Vanishing Viscosity for Hamilton–Jacobi Equations and Applications to Operator Splitting

31. *Applied and Computational Mathematics Seminar*, Auburn University, Math Dept. Apr 20, 2026
Vanishing Viscosity for Hamilton–Jacobi Equations and Applications to Operator Splitting

30. *Special Session on Interaction between Geometric and Analytic Aspects in PDEs* Apr 18, 2026
2026 AMS Spring Central Sectional Meeting, North Dakota State University, Fargo, ND
Vanishing Viscosity for Hamilton–Jacobi Equations and Applications to Operator Splitting

29. *Baylor Applied Math Seminar*, Baylor University, Department of Mathematics Mar 26, 2026
Vanishing Viscosity for Hamilton–Jacobi Equations: overview and recent developments

28. PDE Geometric Analysis seminar, University of Wisconsin - Madison Feb 16, 2026
Rates of Vanishing Viscosity for Hamilton–Jacobi Equations with State Constraints

27. *Special Session on Advances and challenges in hyperbolic conservation laws* Oct 25, 2025
Fall 2025 AMS Eastern Virtual Sectional Meeting, Virtual
Vanishing discount limits for first-order fully nonlinear Hamilton-Jacobi equations on noncompact domains

26. *Special Session on Recent Advances in Numerical Methods for ODE/PDEs and Their Applications* Oct 04-05, 2025
Fall 2025 AMS Southeastern Sectional Meeting, Tulane University, New Orleans, LA
Hamilton–Jacobi equations with state constraints

25. (Online) Mitake-Tran PDE reading seminar, UW-Madison Oct 01, 2025
Rates of Vanishing Viscosity for Hamilton–Jacobi Equations with State Constraints

24. *Baylor Applied Math Seminar*, Baylor University, Department of Mathematics Sep 08, 2025
Rates of Vanishing Viscosity for Hamilton–Jacobi Equations with State Constraints

23. *Workshop on Partial Differential Equations, Mathematical Physics and Numerics 2025* Aug 16-17, 2025
Texas A&M University, Department of Mathematics
Rates of Vanishing Viscosity for Hamilton–Jacobi Equations with State Constraints

22. (Withdrawn) Workshop: "*Recent progress in Hamilton–Jacobi equations and related topics*" Jun 2-6, 2025
Nanjing, China

21. PDEs Seminar, University of Tennessee - Knoxville Nov 07, 2024
Convergence Rate for 1D *Quasi-Periodic Homogenization of Hamilton-Jacobi Equations*

20. Colloquium, Minnesota State University - Mankato Oct 29, 2024
Regularity and Vanishing Viscosity Rates in the Ergodic Problem

19. Analysis Seminar, University of Maryland, College Park Oct 24, 2024
Convergence Rate for 1D *Quasi-Periodic Homogenization of Hamilton-Jacobi Equations*

18. Analysis and PDE Seminar, Michigan State University Oct 16, 2024
Regularity and Vanishing Viscosity Rates in the Ergodic Problem

17. (Online) Analysis Seminar, Texas Tech University Oct 14, 2024
Remarks on the well-posedness of Viscosity Solutions for the One-Phase Muskat Problem

16. Mini-workshop: Summer School in PDEs and Applications 2024 Jul 27, 2024
Vietnam Institute for Advanced Study in Mathematics (VIASM) and Saigon University (SGU)
Remarks on the well-posedness of Viscosity Solutions for the One-Phase Muskat Problem

15. (Online) Virtual Student PDEs Seminar, UW–Madison May 30, 2024
Convergence Rate for 1D *Quasi-Periodic Homogenization of Hamilton-Jacobi Equations*

14. PDEs Seminar, The Ohio State University Apr 09, 2024
Properties of the effective Hamiltonian and homogenization of the Hamilton-Jacobi equation

13. (Online) Seminars on Analysis and Stochastic Analysis, Auburn University — Mar 27, 2024
Properties of the effective Hamiltonian and homogenization of the Hamilton-Jacobi equation

12. (Online) Early Career Math Colloquium, University of Arizona — Mar 21, 2024
Properties of the effective Hamiltonian and homogenization of the Hamilton-Jacobi equation

11. SIAM Great Lakes Meeting (SIAMGL) 2023, Michigan State University — Oct 15, 2023
Minisymposium: Nonlinear PDEs & Optimal Transport with Applications
Generalized convergence of solutions to contact Hamilton–Jacobi equations with state-constraint

10. Differential Equations and Nonlinear Analysis Seminar, North Carolina State University — Nov 09, 2023
Generalized Convergence of Hamilton–Jacobi equations with state-constraint

9. Applied Analysis Seminar, Stinghua University, China — Aug 03, 2023
Generalized convergence of solutions to contact Hamilton–Jacobi equations with state-constraint

8. Analysis Seminar, University of Science, VNU-HCMC — Jun 20, 2023
Generalized convergence of solutions to contact Hamilton–Jacobi equations with state-constraint

7. Madison PDEs Conference, UW-Madison — May 15-19, 2023
Generalized convergence of solutions to contact Hamilton–Jacobi equations with state-constraint

6. (Online) Academy of Mathematics and Systems Science, Chinese Academy of Science — Apr 19, 2023
The regularity with respect to domains of the additive eigenvalues of superquadratic Hamilton–Jacobi equation

5. (Online) Academy of Mathematics and Systems Science, Chinese Academy of Science — Apr 26, 2023
Remarks on the rate of convergence for vanishing viscosity procedure

4. Applied Math Seminar, University of North Carolina - Charlotte — Sep 24, 2021
Remarks on the rate of convergence for vanishing viscosity procedure

3. (Online) PDE Real Analysis Seminar — Oct 27, 2020
Graduate School of Mathematical Sciences, The University of Tokyo
Vanishing discount problems for Hamilton–Jacobi equations on changing domains

2. PDEs and Geometric Analysis Seminar, UW-Madison — Sep 23, 2019
State constraints static Hamilton–Jacobi equations in nested domains

1. 11th Summer Meeting Conference, University of Science, VNU-HCMC — Jul 30, 2019
State constraints static Hamilton–Jacobi equations in nested domains

Selected Contributed Talks & Posters

8. Poster: IMSI's workshop Mathematical Modeling of Biological Interfacial Phenomena — Dec 09-13, 2024
The Institute for Mathematical and Statistical Innovation, University of Chicago
Convergence rate for quasi periodic homogenization of Hamilton–Jacobi equations

7. Boston University/Keio University/Stinghua University Workshop 2024 — Jun 01, 2024
on Differential Equations, Dynamical Systems and Applied Mathematics
Rate of convergence for quasi-periodic homogenization of Hamilton–Jacobi equation and application

6. Analysis and PDE Seminar, Michigan State University — Apr 17, 2024
Convergence Rate for 1D Quasi-Periodic Homogenization of Hamilton-Jacobi Equations

5. 88th Midwest PDEs Seminar, The Ohio State University — Apr 26-28, 2024
Convergence Rate for 1D *Quasi-Periodic Homogenization of Hamilton-Jacobi Equations*

4. Poster: 8th Annual Scholar Showcase — Apr 06, 2024
Office of International Students and Scholars, Michigan State University
On the regularity of the stochastic Hamiltonian

3. Concentration week on Geometry and Analysis, University of Texas A&M — Jul 29, 2022
Remarks on rate of vanishing viscosity for Hamilton–Jacobi equations

2. Geometric and Harmonic Analysis 2019, University of Connecticut — Mar 30, 2019
   Some recent works on homogenization of Hamilton–Jacobi equations

1. Poster: CNA Workshop 2019: Mathematical Models for Pattern formation — Mar 08, 2019
   Carnegie Mellon University
   Homogenization of Hamilton–Jacobi equations: Rate of convergence

Selected Conferences, Workshops Attended & Research Visits

8. AMS 2026 Fall Western Sectional Meeting, Arizona State University, Tempe, Tempe, AZ — Nov 7-8, 2026

7. *Workshop: Applied Mathematics Workshop 2026* — July 20-21, 2026
   University of Economics Ho Chi Minh City (UEH), Vietnam

6. 105th Annual Meeting of the Texas Section of the Mathematical Association of America — Mar 27–28, 2026

5. AMS 2025 Fall Southeastern Sectional Meeting, Tulane University — Oct 01-03, 2025

4. *Workshop on Partial Differential Equations, Mathematical Physics and Numerics 2025* — Aug 16-17, 2025
   Department of Mathematics, Texas A&M University

3. American Institute of Mathematics — Apr 14-18, 2025
   AIM workshop: *Integro-differential equations in many-particle interacting systems*
   Richard N. Merkin Center for Pure and Applied Mathematics, Caltech

2. AMS 2025 Spring Central Sectional Meeting, University of Kansas — Mar 29-30, 2025

1. IMSI's workshop *Mathematical Modeling of Biological Interfacial Phenomena* — Dec 09-13, 2024
   The Institute for Mathematical and Statistical Innovation, the University of Chicago